1. Quality management

1.1. Quality Control and Quality Assurance

What is Quality?

Quality Product: A good or service that meets customers' expectations and therefore fulfils its intended purpose.

What can "quality" mean?

- It can be meeting customer's expectation. Quality can be whether the product meets customer expectations at its price point.
- It can be of the highest standards or mediocre, therefore it is varied. Some products, like aircraft parts, require extreme precision, while others, like clothing, do not. Overengineering can make products too costly.
- Low-cost products can still be high quality if they perform as expected, while expensive ones can be low quality if they fail to meet expectations.
- Quality is not absolute; it depends on the product, its price, and consumer perceptions.

Quality Standards:

Quality standards refer to the minimum acceptable production or service levels based on customer expectations.

They apply to both goods and services:

- Goods: Examples consisting the reliability of cars or the durability of clothing.
- Services: Quality can be measured by factors like:
  - Speed and manner of customer service (e.g., answering calls).
  - Waiting times (e.g., in banks or stores).
  - Accuracy (e.g., error rates in financial transactions).
  - Expertise (e.g., quality of financial advice).

Importance of Quality:

- Satisfied customers are more likely to be loyal to the brand and recommend the brand.
- Fewer complaints, replacements, and compensation expenses; thus it lowers cost.
- Reliable products have a long life cycle and remain in demand for longer.
- A strong quality reputation reduces the need for heavy marketing.
- Quality products can justify businesses charging premium prices.

Quality and Business Competitiveness:

- Consistent quality is often more effective than lowering prices.
- High quality builds brand reputation, allowing businesses to charge premium prices.
- As incomes grow, customers demand higher quality, making brand differentiation essential.
- Businesses must balance quality improvements with potential gains in competitiveness.

Quality Control

Quality control ensures product or service quality through inspection and checking. It typically occurs at the end of production or during service delivery.

Methods of Quality Control:

- Prevention: Designing quality into the product will reduce the need for later corrections.
- Inspection: Checking finished products, though costly and requiring skilled inspectors.
- Correction & Improvement: Fixing faults and improving processes to prevent future defects.

Impact of Quality Control:

Positive Impact	Negative Impact
It helps maintain product standards and meet customer expectations.	Inspectors focus on finding faults, which may demotivate workers.
Reliable quality boosts customer trust and loyalty, boosting brand reputation.	Employees may feel inspectors are only there to criticize.
Identifies and removes faulty products before reaching customers.	Inspectors may lose motivation, reducing efficiency.
Fewer defective products mean lower returns and legal issues.	Issues may go unnoticed until later production stages. It increases replacement cost and removal cost.
Identifying recurring issues helps refine production methods, boosting efficiency.	Employees might only rely on inspectors, reducing accountability for quality.

In modern times, due to new methods being innovated and many flaws of quality control, many businesses are quitting usage of quality control.

Quality Assurance

A system of agreeing and meeting quality standards at each stage of production to ensure consumer satisfaction.

Methods of Quality Assurance:

- It puts highest priority on prevention as it designs products for easy, fault-free manufacturing to get quality right the first time.
- Encouraging workers to aim for zero defects to reduce errors and costly rework.

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- Setting and maintaining targets from product design to final delivery, boosting quality standard.
- It leads to checking raw materials and components upon arrival to avoid defects later.
- Ensuring high-quality delivery, service, and after-sales support.
- Workers check predetermined quality standards at every stage of vehicle factories like Toyota.
- Telephone banking organizations setting strict response times and service quality standards for customer requests.

Impact of Quality Assurance:

Positive Impact	Negative Impact
Employees take responsibility for quality, leading to job enrichment.	Training employees and setting quality standards require significant investment.
Less reliance on final inspections and reworking of faulty products, saving capital that would've been required for inspection.	Continuous monitoring and process improvements is time consuming and can slow down production.
Problems can be traced back to specific production stages.	Workers may find self-checking stressful or resist additional responsibilities.
Preventing defects reduces wastage and quality-related expenses.	Ensuring quality at every stage requires strict coordination and oversight.
Quality products over time will help the firm gain international recognition and improves brand reputation, by achieving ISO 9000-like quality certifications.	Adapting to new quality systems may temporarily reduce efficiency.

ISO 9000:

an internationally recognised standard that acknowledges the existence of a quality assurance method that meets certain conditions.

Requirements to obtain the ISO 9000 certification:

- Regular training and performance evaluation to maintain quality standards.
- Ensuring raw materials and components meet required standards.
- Applying consistent quality management, including after-sales service.
- Clear procedures for identifying, correcting, and preventing quality failures.
- Achieving ISO 9000 enhances credibility, efficiency, and customer satisfaction.

Total Quality Management (TQM)

An approach to quality that aims to involve all employees in quality improvement.

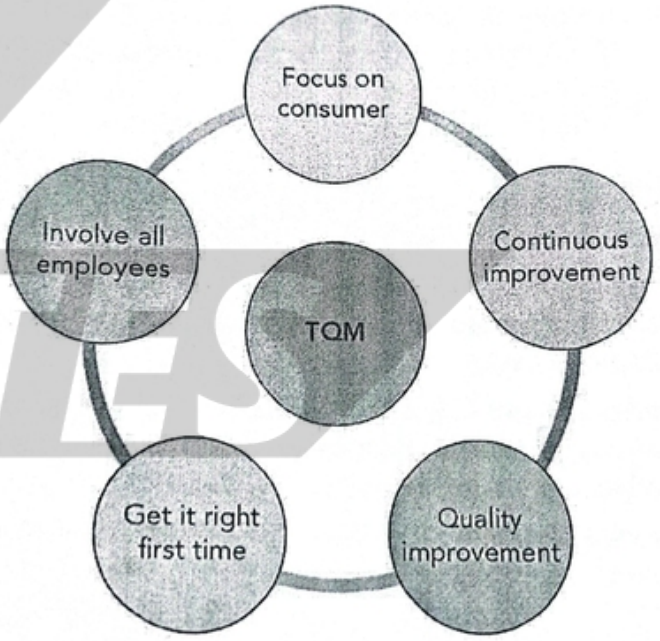
Internal customers: people within the organisation who depend upon the quality of work being done by others.

Zero defects: a way of thinking to ensure that products meet customer expectations every time.

Impact of TQM:

Requirements	Positive Impact	Negative Impact
Senior management must support and promote a culture of quality.	It encourages a zero defects approach, reducing waste and increasing reliability, improving product's quality	Requires large investment in training, new processes, and cultural change.
Workers must be trained and empowered to take responsibility for quality.	It reduces defects, rework, and the need for quality inspections, lowering overall costs.	Continuous improvement and employee involvement is time consuming can slow down operations.
A focus on ongoing evaluation and enhancement of processes to ensure consistent quality. It can be considered absolute Kaizen as it relies on ideology of constant improvement.	Consistently high-quality products enhance brand reputation and customer loyalty. Thus higher customer satisfaction is achieved.	Employees may resist additional responsibilities or struggle to adapt to new quality standards. It might even demotivate or stress out workers.

Visual example of TQM:



1.2. Benchmarking

Comparing a business against the performance standards of the best businesses in the same industry.

Benchmark Performance Indicators (BPI): the areas of business performance that are measured and compared with other businesses. It is also known as Key Performance Indicator (KPI).

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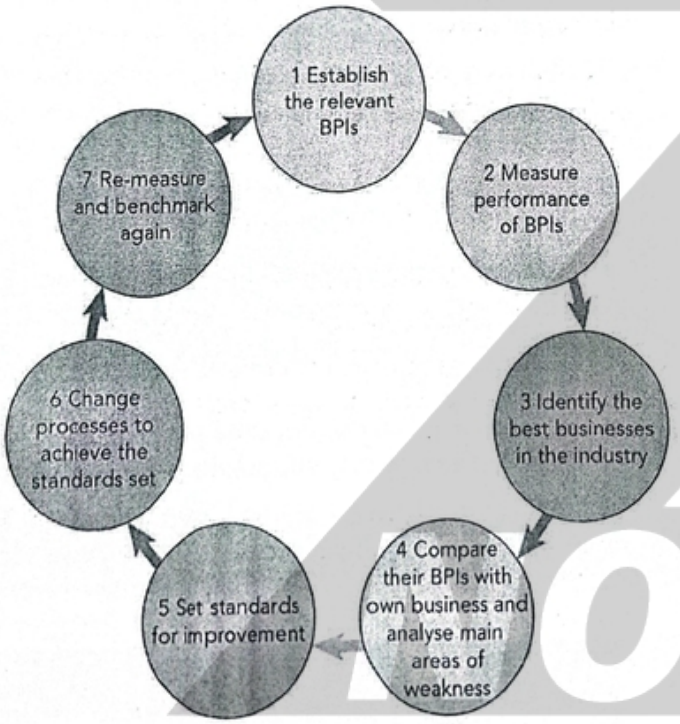
Methods of Benchmarking Process:

- **Identify Benchmark Performance Indicators (BPIs):** Determine what factors (e.g., reliability, delivery speed) are most important to customers.
- **Measure Current Performance:** Gather data on how well the business performs in these areas.
- **Identify Industry Leaders:** Find the best-performing businesses for comparison.
- **Analyze Performance Gaps:** Use data to identify weaknesses and areas for improvement.
- **Set Improvement Standards:** Establish new performance targets based on the best in the industry.
- **Implement Changes:** Enhancing processes to meet or exceed benchmark standards.
- **Re-measure & Continuously Improve:** Regularly assess progress and refine processes to maintain competitiveness.

Visual Example of Benchmarking:



Benefits and Drawbacks of Benchmarking:

Benefits of Benchmarking	Drawbacks of Benchmarking
It helps businesses find and fix weaknesses more effectively.	Obtaining accurate and up-to-date industry information can be challenging.
It improves International Competitiveness – Enables businesses to compete globally, boosting customer satisfaction.	Copying other businesses may limit innovation and unique strategies.
It allows businesses to adopt successful practices from other industries.	The expenses of benchmarking may outweigh the benefits if improvements are minimal and it doesn't result in net profit.
Boosting worker's involvement and motivation by engaging workers in quality improvement.	N/A

Evaluation of Quality Management:

Quality is essential for all businesses, not just manufacturers. It plays a crucial role in customer satisfaction, cost reduction, and competitiveness.

- Both manufacturing and service sectors (e.g., tourism, insurance) must prioritize quality.
- Lower defect rates; reduce waste, rework, and compensation costs.
- Meeting customer expectations encourages repeat business- sign of customer loyalty.
- Employee involvement in quality initiatives improves morale/motivation and engagement.
- Quality products and services boost brand reputation and sales, leading to growth of a company.

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